efforts should be aimed at validating and improving the models used, but in applied studies we need to make a reasonable choice between the methods available at present. With the end-user of our results in mind, it seems pragmatic (and feasible) to extend our analyses of spinal loads (in asymmetric tasks) to the level of forces acting on the intervertebral joints. In view of the assumptions involved in the estimation of these forces, it is our contention that the best report of results also contains information on the underlying net moments and (asymmetry) of posture. Evidently, presenting these net moments in the axis system of the spinal forces will facilitate interpretation. When forces on L5-S1 are reported this axis system would closely follow a conventional pelvis axis system, but it would be inclined with respect to the pelvis axis system, since the latter is usually defined as vertical in upright stance. For reporting of posture a thorax-based reference system as used by Gagnon et al. is needed, since the position of the thorax with respect to the pelvis largely defines the geometry of the musculature.

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Reply by authors

Joint coordinate systems of axes for coherence in reporting kinematic and kinetic data

It is with great interest that we read the letter to the editor by Dr. van Dieën and Dr. Kingma about our paper 'Strategies of load tilts and shoulders positioning in asymmetrical lifting. A concomitant evaluation of the reference systems of axes'. The authors seem to discourage the users from adopting a joint system of axes on the ground that it is not straightforward and somewhat complicated because it is a non-orthogonal system. We do not agree with this position as our paper is addressed to scholars in biomechanics; some researchers in biomechanics may not be familiar with this approach, but the developments have been nicely presented by Fujie et al. [1], and the mathematics can be grasped by researchers with a solid formation in mechanics. They have described mathematically the six-degree-of-freedom forces and moments with respect to a commonly used knee joint coordinate system (namely the joint system described by Grood and Suntay [2]) by performing a $[6 \times 6]$ Jacobian matrix, and they stressed that it is critical to describe the kinematic and kinetic information with respect to the same coordinate system. At this point, it is important to mention that the problem addressed here refers to a 3D representation of moments not different from the problem of 3D representation of angles. We therefore support the view of Fujie et al., especially in studies dealing with the description of asymmetries of posture and efforts. However, this does not eliminate the problem of selecting the axes, one on each segment forming the joint.

Drs van Dieën and Kingma suggested to go one step further and quantify back load in terms of the forces acting on the spine rather than only net moments using one of the two vertebrae; they further suggested the use of the thorax to represent the geometry of the musculature. We believe that to pass from a model representing net moments to a model representing internal forces does not per se represent a solution, since it does not remove the problem of defining axes in any case. Moreover, even if we agree that the use of a small element characterised by L5 could probably reflect better what is actually occurring at this level as compared to larger segments, the sensitivity to torsion would probably be lost due to the very narrow distances (heights) between markers required to define the longitudinal axis of this small element and hence, torsion. As Kingma et al. [3] pointed out, our trunk reference system will not be sensitive to torsion in certain postures. We have also demonstrated at the ISB conference in 1999 in Calgary that the reciprocal can also be ap-

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plied to the pelvis system. We believe that there are some drawbacks in using segmental reference systems (whether the trunk, the pelvis or the vertebrae L5 or S1) when one is concerned with knowing what is occurring at the joint itself. We then favour the adoption of a joint coordinate system of axes; we also favour the use of the same joint system of axes for compatibility in reporting kinematic and kinetic data, i.e., asymmetry of posture (joint orientation) and asymmetry of efforts (moments).

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